

AI Job Impact & Market Demand Analysis

(Simulated Data) - Project Documentation

Author	Santosh Chaurasia
Project type	Practice project
Tools	Python, SQL (SQL Server), Excel, Power BI
Dataset	Simulated data, 1,500 rows, 2021 to 2026

Note: The data in this project is simulated. I made it with Python to practice the full analysis steps. It does not show the real job market, so please do not take the results as real facts.

1. About the Project

In this project I wanted to practice a complete data analysis from start to end. I made a dataset of 1,500 rows about 6 job roles from 2021 to 2026 and checked how vacancies, salaries, skills and degree requirements change after AI.

I treated 2021-2022 as **before AI** and 2023-2026 as **after AI**. I did the analysis in four tools (Python, SQL, Excel and Power BI) and checked that the totals match in all of them.

The six roles are Web Developer, Data Analyst, Software Engineer, Data Entry Operator, Manual QA Tester and AI Prompt Engineer.

2. Goals

- Compare vacancies and salaries before and after AI.
- Find which roles and skills are growing or declining.
- Check if degree importance changes and where a portfolio matters more.
- See how hiring difficulty is related to salary.
- Find the salary growth of Data Analysts from 2021 to 2026.
- Show the results in an Excel dashboard and a Power BI dashboard.

3. Dataset

The file `ai_job_impact_data.csv` has 1,500 rows and 9 columns. I checked it with `df.info()` and there are no missing values.

Column	Meaning
Year	2021 to 2026
Job_Role	One of the 6 roles
Vacancies	Number of openings (10 to 249)
Hiring_Difficulty	Easy, Medium or Hard
Top_Required_Skill	Main skill for that role
Skill_Demand_Status	Stable, Growing, High Growth or Declining
Degree_Importance	Low, Medium or High
Portfolio_Weightage_Percentage	How much a portfolio matters in hiring (20 to 94)
Avg_Salary_LPA	Average salary in lakhs per year (2.0 to 14.98)

How I made the data: I used NumPy random choices with seed 42 in the notebook `AI_Job_Impact_Data_Generation.ipynb`. I gave different ranges for each role and for before and after AI. For example, after 2022 Data Entry Operator and Manual QA Tester get low vacancies and a "Declining" skill, and AI Prompt Engineer gets high salaries. Because I set these rules myself, the analysis only confirms them, it does not discover them.

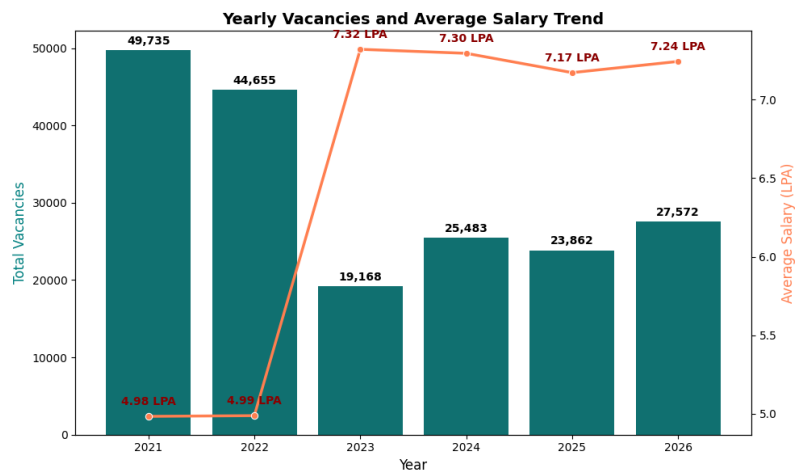
AI Prompt Engineer is a new role, so it has no rows for 2021 and 2022.

4. Steps I Followed

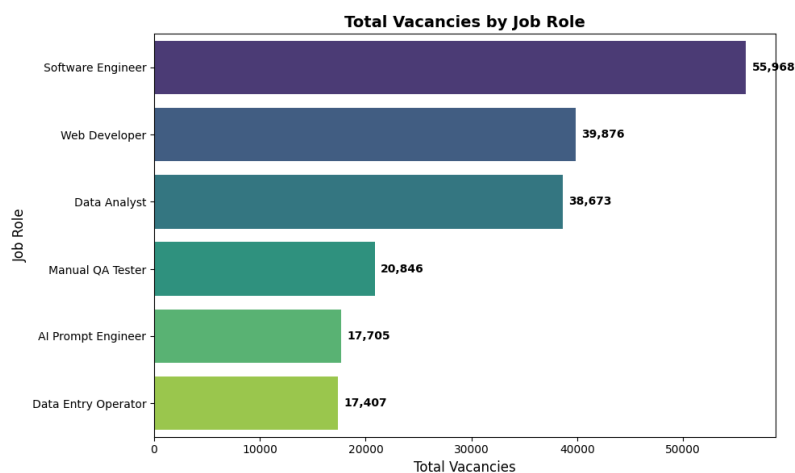
- Generated the dataset in Python and saved it as a CSV file.
- Loaded the CSV in Python, checked it and made 3 charts.
- Imported the CSV into SQL Server and wrote 10 queries.
- Made pivot tables, pivot charts and slicers in Excel.
- Made a one-page dashboard in Power BI.
- Checked that the totals match everywhere (total vacancies = 190,475).

5. Python

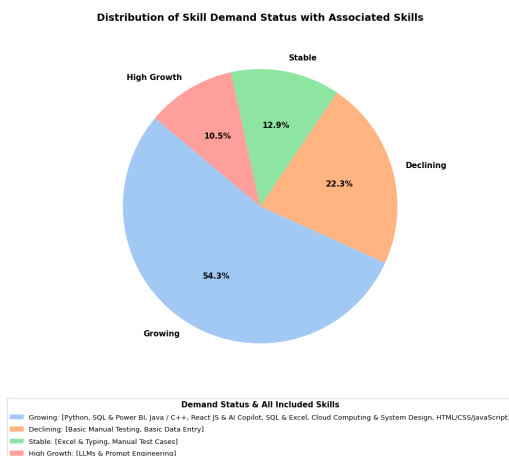
I ran `AI_Impact_On_Job_Python_Analysis.ipynb` in Google Colab. I loaded the data with Pandas, checked data types and missing values, and grouped the data by year and by job role. Then I made three charts.



Total vacancies fall from 49,735 in 2021 to 19,168 in 2023, while the average salary jumps from about 4.99 to 7.32 LPA.



Software Engineer has the most vacancies (55,968), partly because it also has the most rows. AI Prompt Engineer has the fewest (17,705) because it only exists from 2023.



Of the 1,500 rows: 815 Growing, 335 Declining, 193 Stable and 157 High Growth.

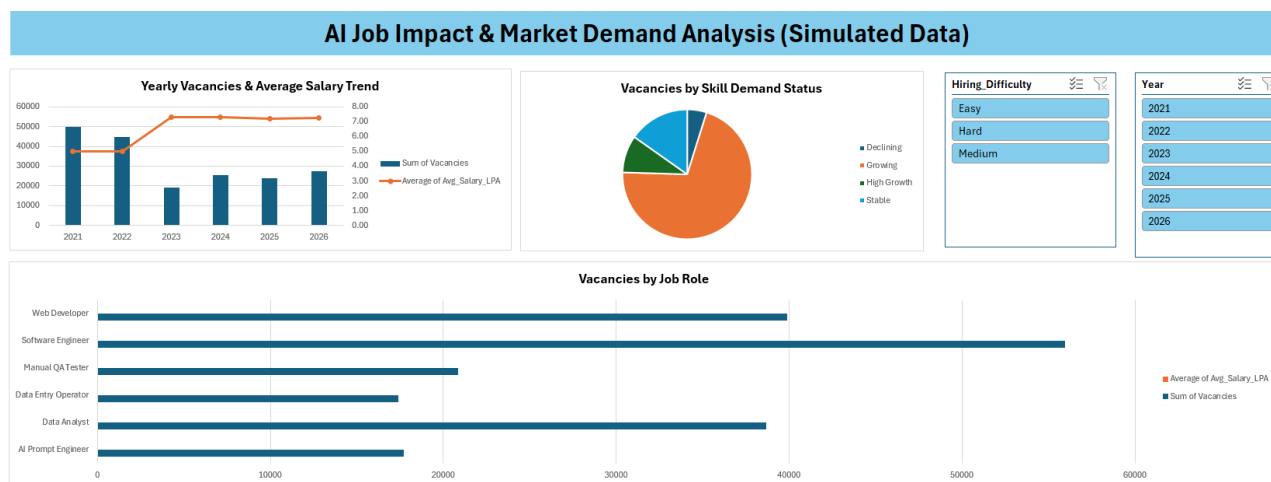
6. SQL

I wrote 10 queries in SQL Server on the table `ai_job_impact`. When I average whole-number columns I multiply by 1.0 so SQL Server does not round the answer. In Query 3 I used a CTE with `ROW_NUMBER()` to rank the roles.

#	Query	Result
1	Year-on-Year Job Market Summary	Total vacancies: 49,735 (2021), 44,655 (2022), 19,168 (2023), 25,483 (2024), 23,862 (2025), 27,572 (2026). Average salary goes from 4.98 to 7.24 LPA.
2	Before vs After AI	Average vacancies per row fall for roles in both periods (Data Analyst 202 to 142). Salary rises for tech roles (Data Analyst 6.00 to 8.93 LPA).
3	Highest Paying Roles	Before: Web Developer 6.04, Software Engineer 6.02, Data Analyst 6.00. After: AI Prompt Engineer 11.33, Software Engineer 9.15, Web Developer 9.02.
4	Declining Skills	Data Entry Operator (Basic Data Entry) and Manual QA Tester (Basic Manual Testing).
5	Degree Requirement Shift	In 2021-2022 no row has Low degree importance. From 2023, Low is the biggest group every year.
6	Hiring Difficulty vs Salary	Medium 7.44 LPA, Hard 6.52 LPA, Easy 4.67 LPA.
7	High Demand Skills (2023 onwards)	LLMs & Prompt Engineering, Python, SQL & Power BI, Cloud Computing & System Design, React JS & AI Copilot.
8	High Portfolio, Low Degree	187 rows have portfolio weightage above 75 with Low degree importance (AI Prompt Engineer 105, Software Engineer 32, Web Developer 29, Data Analyst 21).
9	Data Analyst Salary Growth	6.07 LPA in 2021 to 8.39 LPA in 2026, a growth of 38.18%.
10	Biggest Vacancy Drop (2021 vs 2026)	Data Entry Operator 155 to 26, Manual QA Tester 149 to 28, Software Engineer 199 to 138, Data Analyst 200 to 140, Web Developer 197 to 141 (average per row).

7. Excel Dashboard

The workbook `AI_Job_Impact_Analysis.xlsx` has a data sheet and a dashboard sheet. I made 3 pivot tables (by year, by job role and by skill demand status), 3 pivot charts and 2 slicers (Year and Hiring_Difficulty). The grand totals match the other tools: 190,475 vacancies and 6.47 LPA average salary.

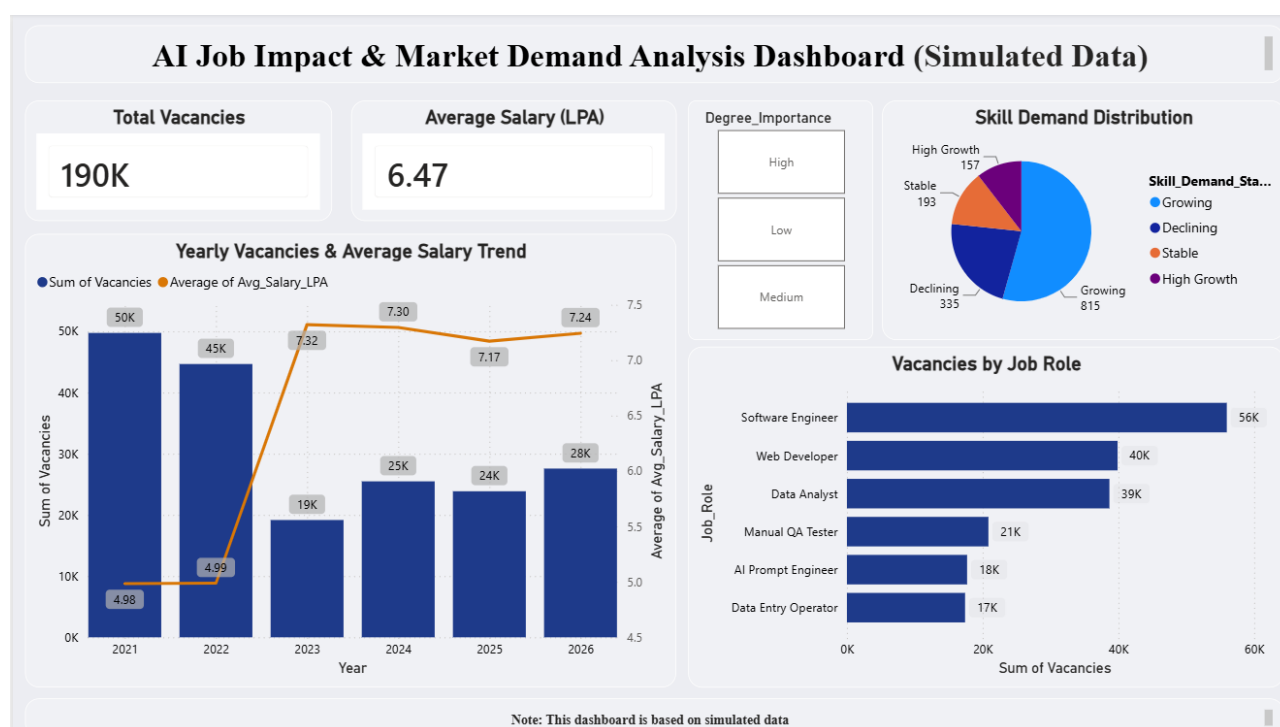


Excel dashboard (simulated data).

The Excel pie chart shows vacancies for each skill status, but the Power BI and Python pie charts count rows, so they are not the same measure.

8. Power BI Dashboard

I made a one-page dashboard with 2 KPI cards (Total Vacancies and Average Salary), a combo chart for yearly vacancies and salary, a pie chart for skill demand, a bar chart for vacancies by job role and a Degree Importance slicer. The title and a footer note say that the data is simulated.



Power BI dashboard (simulated data).

9. What I Found

These results are from the simulated data only.

- Total vacancies are 190,475 and the average salary is 6.47 LPA.
- Vacancies went down after 2022. Average vacancies per row went from about 180 to about 98.
- Salaries went up, from 4.99 LPA before AI to 7.26 LPA after AI.

- Data Entry Operator and Manual QA Tester lost the most vacancies (from about 150 to under 30 per row) and both are marked as declining.
- After 2022, AI Prompt Engineer has the highest salary (11.33 LPA), then Software Engineer (9.15) and Web Developer (9.02).
- Data Analyst salary grew by about 38% (6.07 to 8.39 LPA), while average vacancies per row fell from about 200 to 140.
- Degrees matter less after 2022. Low degree importance is the most common value from 2023.
- Medium hiring difficulty has the highest average salary (7.44), then Hard (6.52), then Easy (4.67).

10. Limitations

- The data is simulated and most patterns come from the rules I wrote, so this project shows the analysis steps and not real job market trends.
- AI Prompt Engineer has no 2021 or 2022 rows, so its growth cannot be checked.
- Years do not have the same number of rows, so averages per row are fairer than yearly totals.
- I did not do any statistical tests, only simple averages and counts.

11. What I Learned

- How to check and group data with Pandas.
- How to write SQL with GROUP BY, CASE, a CTE and ROW_NUMBER().
- How to make pivot tables and slicers in Excel and a one-page dashboard in Power BI.
- That the same numbers should match in every tool.
- That simulated data can only show patterns that were put into it.

12. How to Run

- **Python:** open the notebooks in Google Colab or Jupyter and keep `ai_job_impact_data.csv` in the same folder.
- **SQL:** create the database `ai_job_market_db` in SQL Server, import the CSV into a table named `ai_job_impact` and run the query files.
- **Excel:** open the workbook and go to the dashboard sheet.
- **Power BI:** open the `.pbix` file in Power BI Desktop.

Author

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